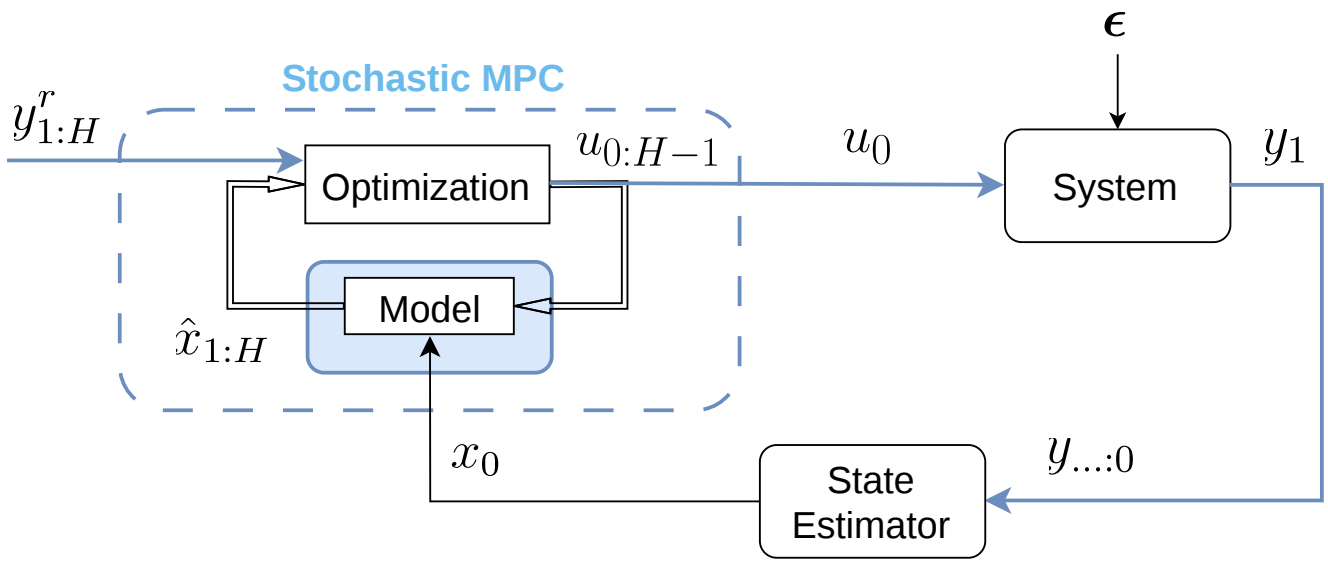
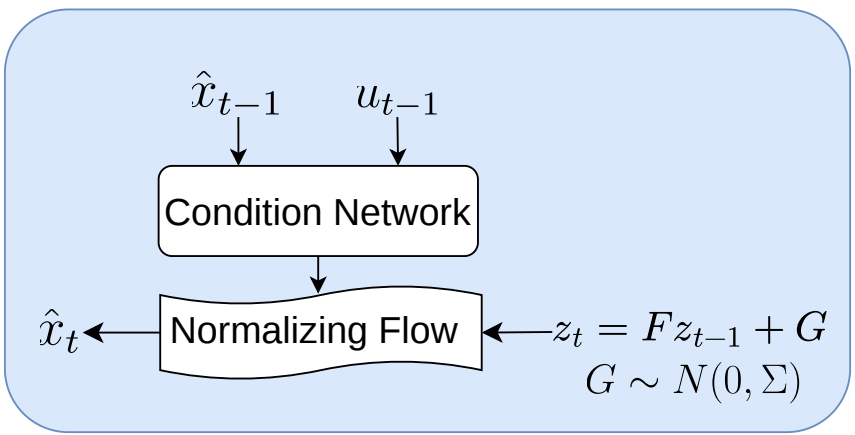
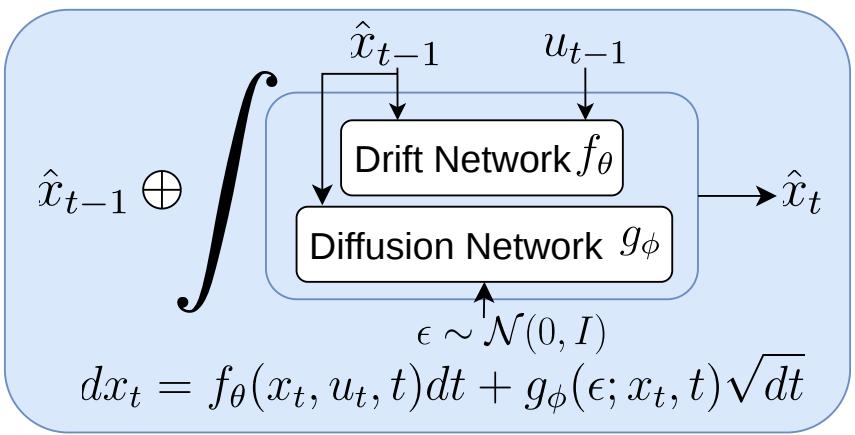

Stochastic MPC
$y^r_{1:H}$
Optimization
$u_{0:H-1}$
$u_0$
$\epsilon$
System
$y_1$
Model
$\hat{x}_{1:H}$
$x_0$
State Estimator
$y_{...:0}$


$\hat{x}_{t-1}$
$u_{t-1}$
Condition Network
$\hat{x}_t$
Normalizing Flow
$z_t = Fz_{t-1} + G$
$G \sim N(0, \Sigma)$


$\hat{x}_{t-1}$
$u_{t-1}$
$\hat{x}_{t-1} \oplus$
Drift Network $f_\theta$
Diffusion Network $g_\phi$
$\hat{x}_t$
$\epsilon \sim \mathcal{N}(0, I)$
$dx_t = f_\theta(x_t, u_t, t)dt + g_\phi(\epsilon; x_t, t)\sqrt{dt}$